I Want It Now:
The Logistics of Delivering Stuff in the Future

Presenter:
Haggai Davis, III

NYU TANDON SCHOOL OF ENGINEERING

C2SMARTER
CONNECTED COMMUNITIES WITH SMART MOBILITY TO EQUITABLY REDUCE CONGESTION

BULET @ NYU
Behavioral Urban Informatics, Logistics, & Transport Lab

Structure:

Outline:

1. Introduction
2. Literature Review
3. Data
4. Methodology
5. Results
6. Conclusion

Structure:

Common Outline:

1. Introduction
2. Literature Review
3. Data
4. Methodology
5. Results
6. Conclusion

Academic:

- Narrow
- Impenetrable
- Impersonal

Narrative Form:

- Honest
- Interesting
- Applicable

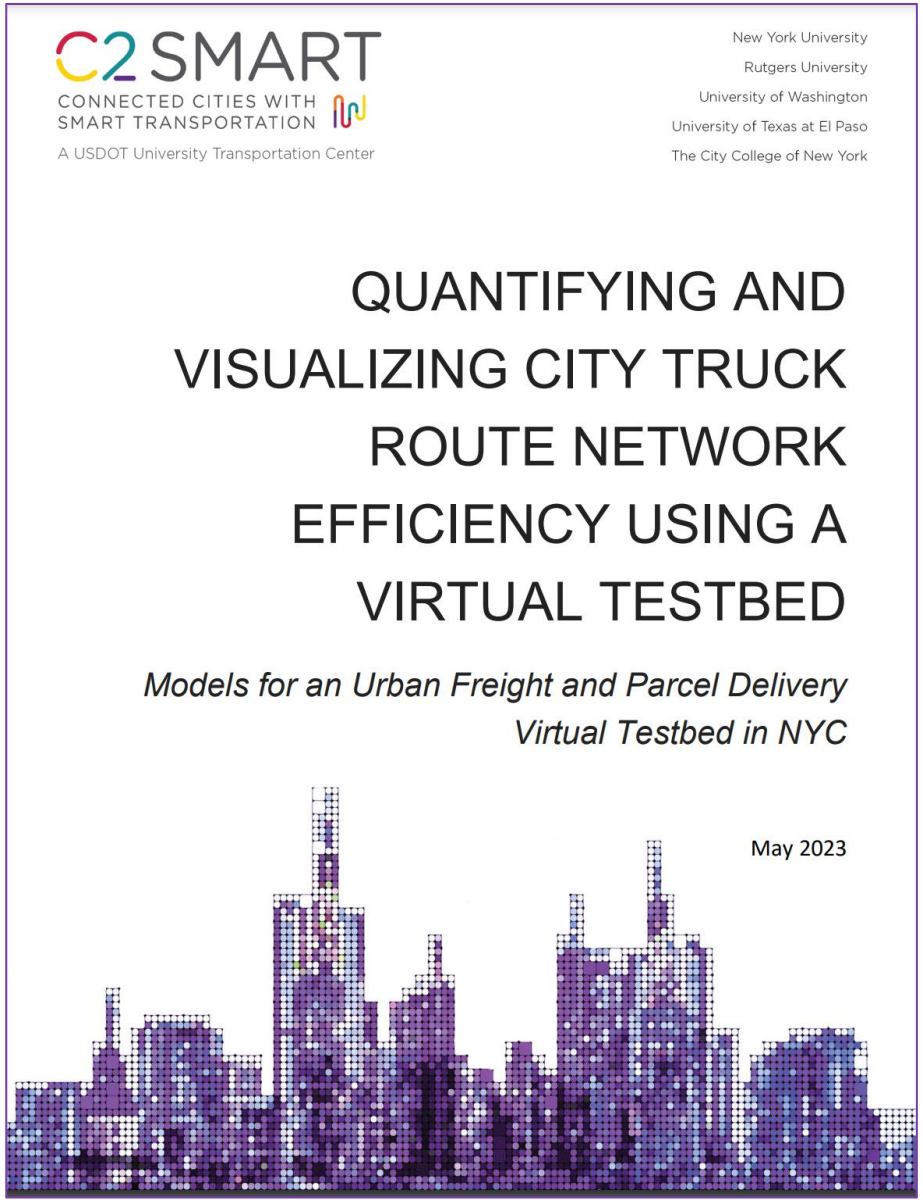
Background:

Wrapped up major project

- 2 years of work
- 8 different researchers
- Multiple presentations
- Academic publication

What did this project do?

What comes next?



C2SMART
CONNECTED CITIES WITH SMART TRANSPORTATION
A USDOT University Transportation Center

New York University
Rutgers University
University of Washington
University of Texas at El Paso
The City College of New York

QUANTIFYING AND VISUALIZING CITY TRUCK ROUTE NETWORK EFFICIENCY USING A VIRTUAL TESTBED

Models for an Urban Freight and Parcel Delivery Virtual Testbed in NYC

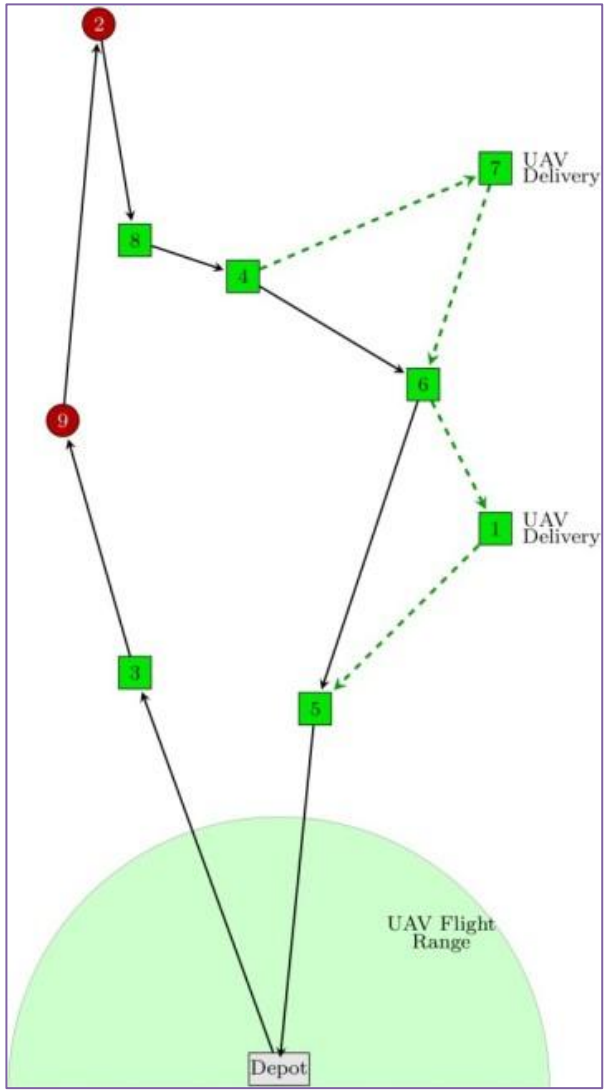
May 2023

Mothership and Flying Sidekick:

- Variant on Traveling Salesman Problem
  - Classic transportation optimization problem
  - Shortest Path + Constraints

- Delivery drone departs from a vehicle already enroute

Source: amazon.com



Source: https://www.chasemurray.com/research/180-2

Refocusing on Automation in NYC:

Benefits:

- Faster?
- Cheaper?
  - Enough to Matter?

Issues:

- Ensure Safety of
  - Package
  - Drone
  - Nearby Pedestrians/Buildings

- Weather Issues
- Compliance with legal guidelines

Source: Bing image creator

Key Idea:

Technology *for the sake of technology*

- A solution looking for a problem

Must clearly identify and understand the problem

Transportation Problems

- Risk profile
- Risk tolerance
- Success vs Failure

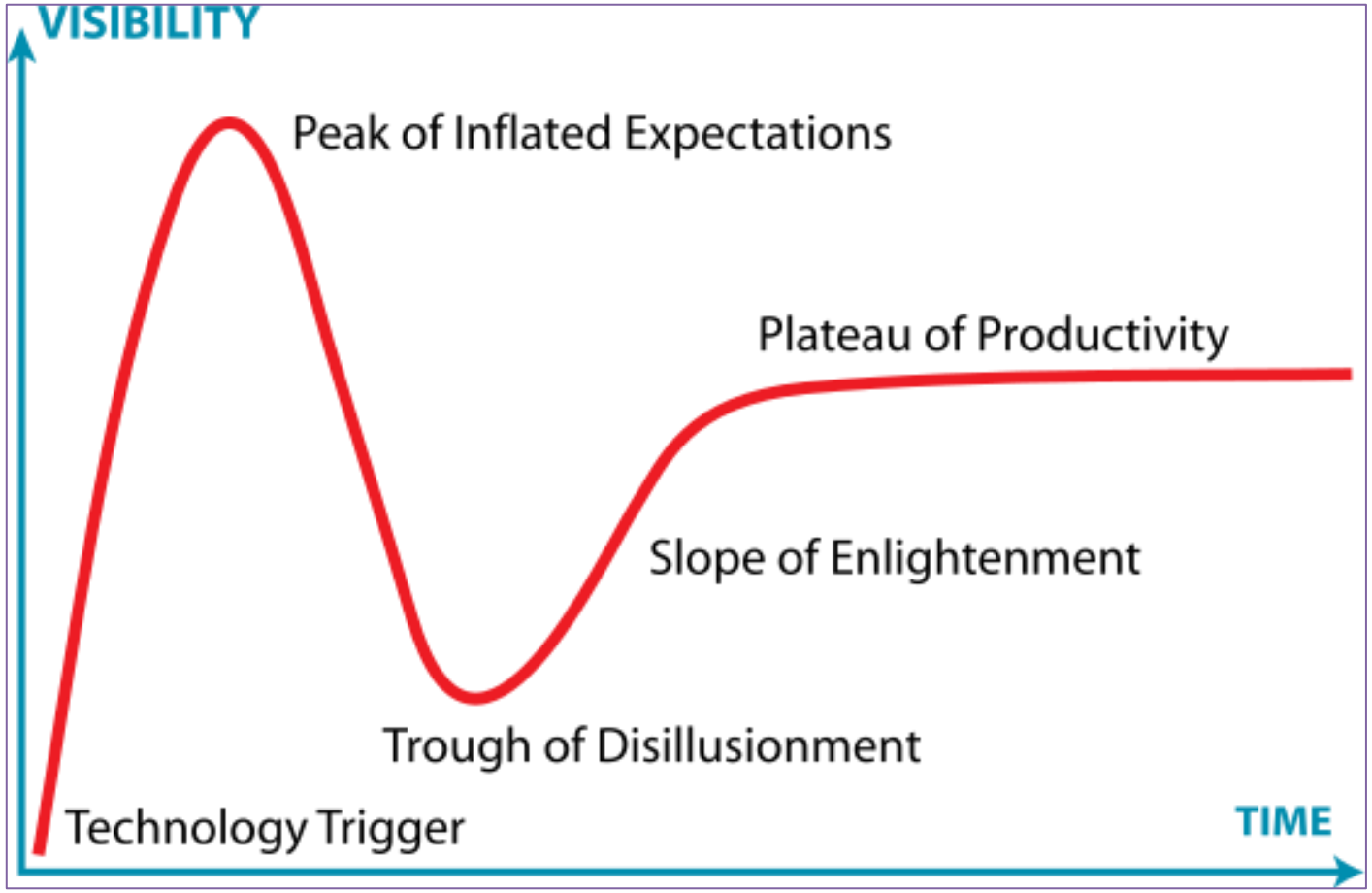
Source: NY Post

What Comes Now?

The Real World

Gartner Hype Cycle:

- Common pattern
- Plenty of Tech Examples:
  - Electric Vehicles
  - 3D Printing
  - Speech Recognition
- Some never recover
  - NFTs and Blockchain?
- Some move faster
  - LLMs
  - Image/Video Generators



Realistic Implementation:

Edges Cases
- Hard
- Whack-a-mole
- Value of human labor

Low Hanging Fruit approach
- Simplify
- Middle Mile

Important Targets:
- Serve currently unmeetable demand
- Augment our current systems

Source: Bing image creator

Realistic Implementation:

Use Case: Rural African Medical Deliveries

Problems Solved:

- Freight is time-sensitive
  - Blood/Vaccines
- Current Infrastructure limitations
  - Poor and non-existence roads

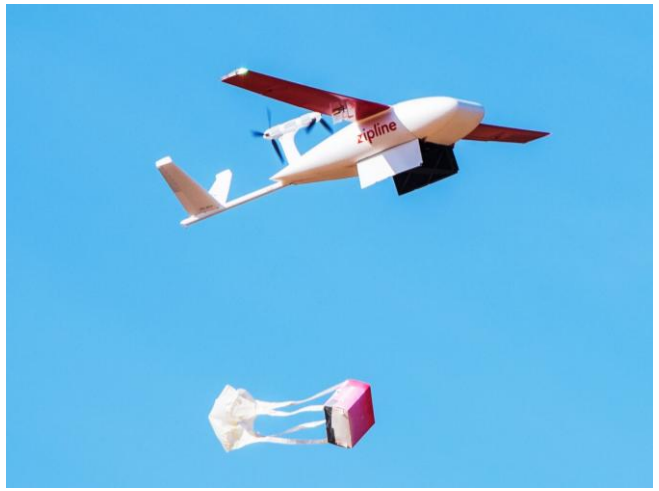
Process:

- Package is parachuted
- Large ground targets

Problems Avoided:

- Drone never slows down
- Drone never nears the ground
- Simplifies everything!

Source: The Guardian

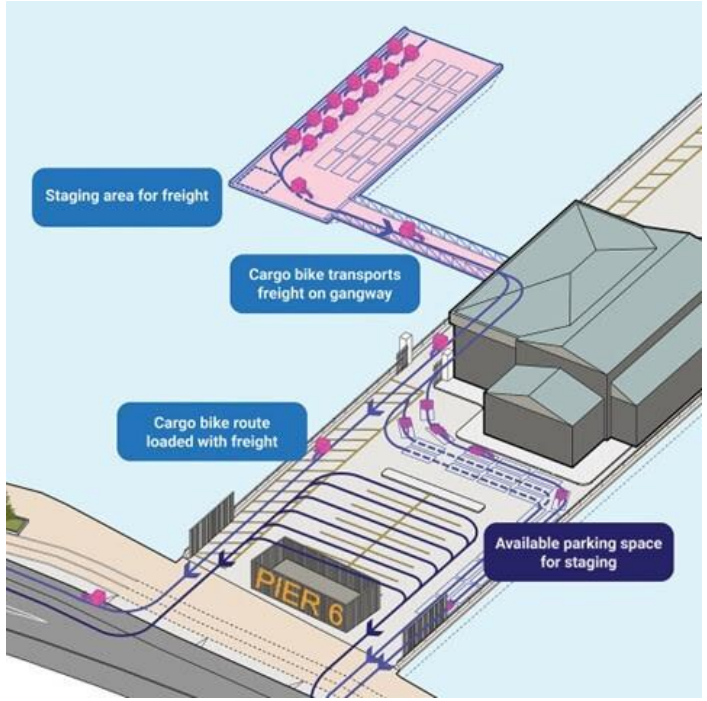


Source: Wired

Realistic Implementation:

Problem: Limited Transportation Supply

- Old Solution: Infrastructure Construction
- New Solution: Rethinking the Limitations
  - Temporal:
    - Autonomous Trucks with Off Hour Delivery
  - Spatial:
    - Shift Freight towards Ports and Barges



Problem: Last Mile is Expensive

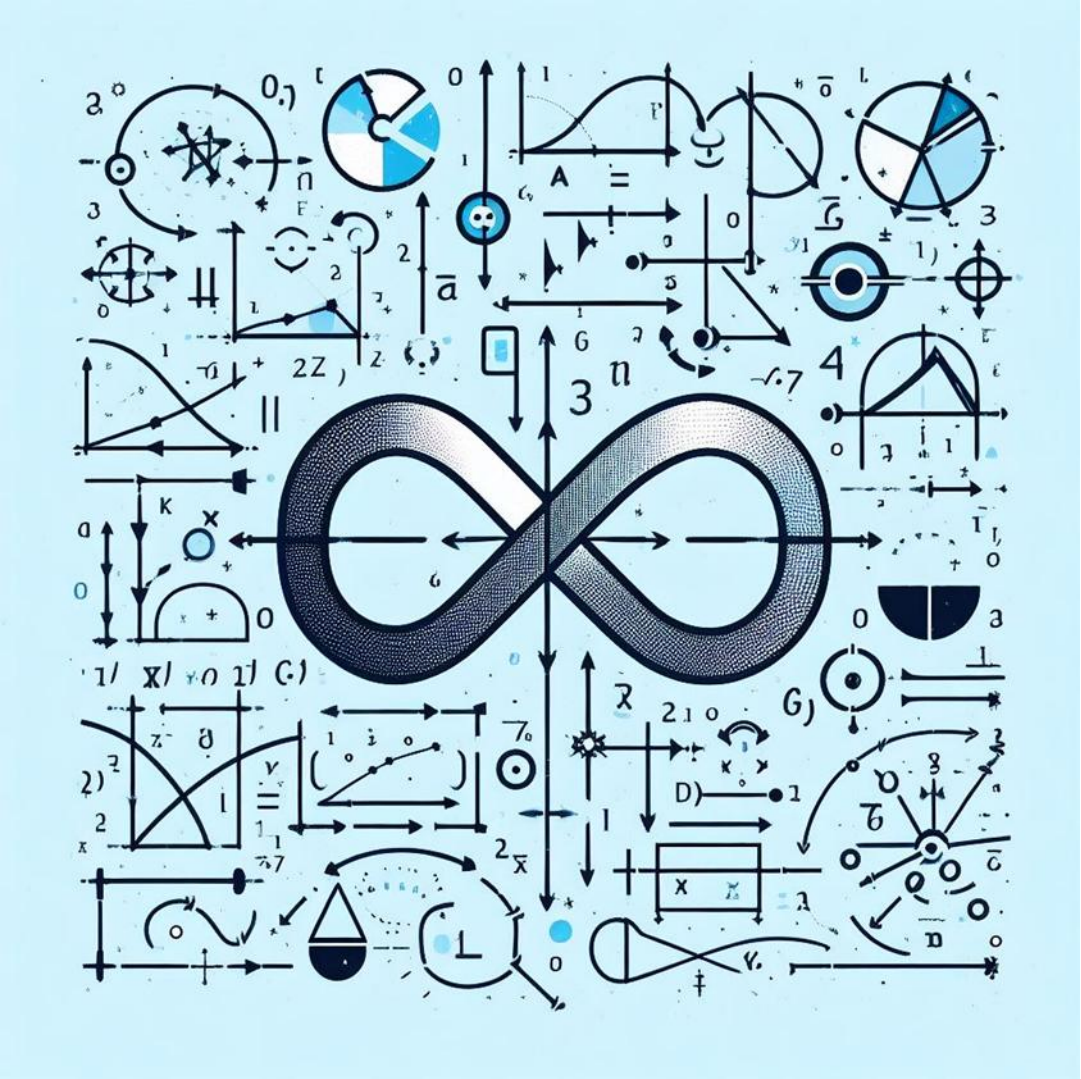
- Old Solution: Warehouse Construction
- New Solution: Mode Shift
  - eCargo Bikes
  - Electric Trucks

Difficulties in Replication:

Obvious Next Question:
Why doesn't everyone get this right all the time?

Answers:

- Good solutions are hard
- The future is unexpected
- Transportation is specific



Source: Bing image creator

Difficulties in Replication:

Thank you!

Haggai Davis III, PhD Candidate
hdd4656@nyu.edu
C2SMART Center
New York University
Tandon School of Engineering

c2smarter.engineering.nyu.edu
c2smart@nyu.edu